

Chapter 12 GUI Basics

Section 12.2 Swing vs. AWT

1. Swing components that don't rely on native GUI are referred to as

- a. lightweight components
- b. heavyweight components
- c. GUI components
- d. non-GUI components

2. _____ are referred to as heavyweight components.

- a. AWT components
- b. Swing components
- c. GUI components
- d. Non-GUI components

Section 12.3 The Java GUI API

3. Which of the following statements are true?

- a. To distinguish new Swing component classes from their AWT counterparts, Swing GUI component classes are named with a prefix J.
- b. All Swing GUI components are lightweight.
- c. A user interface object such as (button, list) can appear in one container.
- d. A container such as JFrame is also a component.
- e. Every instance of Component can be added to a container.

4. Which of the following classes is a heavyweight component?

- a. JButton
- b. JTextField
- c. JPanel
- d. JFrame

5. Which component cannot be added to a container?

- a. JPanel
- b. JButton
- c. JFrame
- d. JComponent

6. Which of the following are subclasses of java.awt.Component?

- a. Container classes
- b. Swing user interface classes
- c. Helper classes such as Color and Font
- d. Layout managers

7. What is best to describe the relationship between a container and a Swing GUI object in the container?

- a. Association
- b. Aggregation
- c. Composition
- d. Inheritance

8. What is best to describe the relationship between a container and a layout manager?

- a. Association
- b. Aggregation
- c. Composition
- d. Inheritance

9. What is best to describe the relationship between JComponent and JButton?

- a. Association
- b. Aggregation
- c. Composition

d. Inheritance

10. What is best to describe the relationship between Component and Color?

- a. Association
- b. Aggregation
- c. Composition
- d. Inheritance

11. What is best to describe the relationship between Component and Font?

- a. Association
- b. Aggregation
- c. Composition
- d. Inheritance

12. Which of the following classes are in the java.awt package?

- a. Color
- b. Font
- c. Component
- d. JFrame
- e. JComponent

Section 12.4 Frames

13. Analyze the following code.

```
import java.awt.*;
import javax.swing.*;

public class Test {
    public static void main(String[] args) {
        Component c = new JButton("OK");
        JFrame frame = new JFrame("My Frame");
        frame.add(c);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setVisible(true);
    }
}
```

- a. You cannot assign a JButton to a variable of java.awt.Component.
- b. You can only add c to a container because c's type is Component.
- c. You cannot add a Swing component directly to a JFrame using add(c) prior to JDK 1.4, but it is OK in JDK 1.5.
- d. You cannot create a JFrame using new JFrame("My Frame").

14. The correct order of the following three statements is _____.

- 1. frame.setLocationRelativeTo(null);
- 2. frame.setSize(100, 200);
- 3. frame.setVisible(true);

- a. 1 2 3
- b. 1 3 2
- c. 2 1 3
- d. 3 2 1

15. Analyze the following code.

```
import java.awt.*;
import javax.swing.*;

public class Test {
    public static void main(String[] args) {
        JFrame frame = new JFrame("My Frame");
```

```

frame.add(new JButton("OK"));
frame.add(new JButton("Cancel"));
frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
frame.setSize(200, 200);
frame.setVisible(true);
}

```

- a. Only button OK is displayed.
- b. Only button Cancel is displayed.
- c. Both button OK and button Cancel are displayed and button OK is displayed on the left side of button Cancel.
- d. Both button OK and button Cancel are displayed and button OK is displayed on the right side of button Cancel.

16. How many frames are displayed?

```
import javax.swing.*;
```

```

public class Test {
    public static void main(String[] args) {
        JFrame f1 = new JFrame("My Frame");
        JFrame f2 = f1;
        JFrame f3 = f2;
        f1.setVisible(true);
        f2.setVisible(true);
        f3.setVisible(true);
    }
}

```

- a. 1.
- b. 2.
- c. 3.
- d. 0.

17. How many frames are displayed?

```
import javax.swing.*;
```

```

public class Test extends JFrame {
    public static void main(String[] args) {
        JFrame f1 = new Test();
        JFrame f2 = new Test();
        JFrame f3 = new Test();
        f1.setVisible(true);
        f2.setVisible(true);
        f3.setVisible(true);
    }
}

```

- a. 1.
- b. 2.
- c. 3.
- d. 0.

18. Which of the following statements is for terminating the program when closing the frame?

- a. frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE)
- b. frame.setDefaultCloseOperation(null)
- c. frame.setDefaultCloseOperation(JFrame.STOP_ON_CLOSE)
- d. frame.setDefaultCloseOperation(JFrame.TERMINATE_ON_CLOSE)

19. Which of the following statements is for placing the frame's upper left corner to (200, 100)?

- a. `frame.setLocation(100, 100)`
- b. `frame.setLocation(100, 200)`
- c. `frame.setLocation(200, 100)`
- d. `frame.setLocation(200, 200)`

24. Analyze the following code:

```
import javax.swing.*;

public class Test1 extends JFrame {
    public Test1() {
        JButton jbt1 = new JButton("OK");
        add(jbt1);
        jbt1 = new JButton("Not OK");
    }

    public static void main(String[] args) {
        JFrame frame = new Test1();
        frame.setSize(300, 300);
        frame.setLocationRelativeTo(null);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setVisible(true);
    }
}
```

- a. The program displays nothing.
- b. The program displays a button with text OK.
- c. The program displays a button with text Not OK.
- d. The program displays a button with text OK and another button with text Not OK.

Section 12.5 Layout Managers

20. What layout manager should you use so that every component occupies the same size in the container?

- a. a `FlowLayout`
- b. a `GridLayout`
- c. a `BorderLayout`
- d. any layout

21. What should you use to position a Button within an application Frame so that the size of the Button is NOT affected by the Frame size?

- a. a `FlowLayout`
- b. a `GridLayout`
- c. the center area of a `BorderLayout`
- d. the East or West area of a `BorderLayout`
- e. the North or South area of a `BorderLayout`

22. Suppose a `JFrame` uses the `GridLayout(2, 2)`. If you add six buttons to the frame, how many columns are displayed?

- a. 1
- b. 2
- c. 3
- d. 4

23. Suppose a `JFrame` uses the `GridLayout(0, 2)`. If you add six buttons to the frame, how many columns are displayed?

- a. 1
- b. 2
- c. 3
- d. 4

24. Suppose a JFrame uses the GridLayout(2, 0). If you add six buttons to the frame, how many columns are displayed?
- a. 1
 - b. 2
 - c. 3
 - d. 4
25. To set a FlowLayout in panel jp, you can use the method _____.
a. jp.setLayout(new FlowLayout());
b. jp.setLayout(new FlowLayout(FlowLayout.CENTER));
c. jp.setLayout(new FlowLayout(FlowLayout.center));
d. jp.setLayout(FlowLayout());
26. The default layout out of a contentPane in a JFrame is _____.
a. FlowLayout
b. GridLayout
c. BorderLayout
d. None

Section 12.6 Using Panels as Sub-Containers

33. The default layout out of a JPanel is _____.
a. FlowLayout
b. GridLayout
c. BorderLayout
d. None
34. To create a JPanel of the BorderLayout, use _____.
a. JPanel p = new JPanel()
b. JPanel p = new JPanel(BorderLayout());
c. JPanel p = new JPanel(new BorderLayout());
d. JPanel p = new JPanel().setLayout(new BorderLayout());
35. To add a component c to a JPanel p, use _____.
a. p.add(c)
b. p.getContentPane(c)
c. p.insert(c)
d. p.append(c)

Section 12.7 The Color Class

27. _____ creates a color object.
a. new Color(0, 0, 0)
b. new Color(0, 266, 0)
c. new Color(255, 255, 255)
d. new Color(1, 2, 3)
28. The method _____ sets the background color to yellow in JFrame f.
a. setBackground(Color.yellow)
b. f.setBackground(Color.YELLOW)
c. f.setBackground(Color.yellow)
d. setBackground(Color.YELLOW)
e. f.setBackGround(Color.yellow)
29. The method _____ sets the foreground color to yellow in JFrame f.
a. setForeground(Color.yellow)
b. f.setForeground(Color.YELLOW)
c. f.setForeground(Color.yellow)
d. setForeground(Color.YELLOW)
e. f.setForeGround(Color.yellow)
30. Which color is the darkest?

- a. `new Color(0, 0, 0)`
- b. `new Color(10, 0, 0)`
- c. `new Color(20, 0, 0)`
- d. `new Color(30, 0, 0)`
- e. `f.setForeground(Color.yellow)`

Section 12.8 The Font Class

31. The method _____ sets the font (Helvetica, 20-point bold) in component C.

- a. `c.setFont(new Font("Helvetica", Font.bold, 20))`
- b. `c.setFont(new Font("helvetica", BOLD, 20))`
- c. `c.setFont(Font("Helvetica", Font.BOLD, 20))`
- d. `c.setFont(new Font("Helvetica", Font.BOLD, 20))`

32. To specify a font to be bold and italic, use the font style value _____

- a. `Font.PLAIN`
- b. `Font.BOLD`
- c. `Font.ITALIC`
- d. `Font.BOLD + Font.ITALIC`

Section 12.9 Common Features of Swing GUI Components

36. Can you use the `setBackground` method to set a back ground color for _____?

- a. `Component`
- b. `Container`
- c. `JComponent`
- d. `JButton`
- e. `JLabel`

37. Can you use the `getWidth` method to get a width for _____?

- a. `Component`
- b. `Container`
- c. `JComponent`
- d. `JButton`
- e. `JLabel`

38. Can you use the `setBorder` method to set a border for _____?

- a. `Component`
- b. `Container`
- c. `JComponent`
- d. `JButton`
- e. `JLabel`

39. Can you use the `setToolTip` method to set a tool tip for _____?

- a. `Component`
- b. `Container`
- c. `JComponent`
- d. `JButton`
- e. `JLabel`

40. You can use methods _____ on any instance of `java.awt.Component`.

- a. `setBackground`
- b. `getBackground`
- c. `getWidth`
- d. `getHeight`
- e. `setBorder`

41. Analyze the following code:

```
import javax.swing.*;

public class Test extends JFrame {
```

```

private JButton jbtOK = new JButton("OK");

public static void main(String[] args) {
    // Create a frame and set its properties
    JFrame frame = new Test();
    frame.setTitle("Logic Error");
    frame.setSize(200, 100);
    frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    frame.setVisible(true);
}

public Test() {
    jbtOK.setToolTipText("This is a button");
    add(new JButton("OK"));
}
}

```

- a. The tool tip text is displayed when you move the mouse on the button.
- b. The tool tip text will be displayed if you replace `add(new JButton("OK"))` with `add(jbtOK)`.
- c. The tool tip text will be displayed if you swap the two lines in the Test constructor.
- d. The tool tip text will be displayed if you replace `add(new JButton("OK"))` with `add(jbtOK = new JButton("OK"))`.

42. You can use methods _____ on any instance of `java.awt.Component`.

- a. `setForeground`
- b. `setBackground`
- c. `setFont`
- d. `getFont`
- e. `setLayout`

43. Show the output of the following code?

```

import javax.swing.*;

public class Test {
    public static void main(String[] args) {
        JButton jbtOK = new JButton("OK");
        System.out.print(jbtOK.isVisible() + ", ");

        JFrame frame = new JFrame();
        System.out.println(frame.isVisible());
    }
}

```

- a. true, true
- b. true, false
- c. false, true
- d. false, false

Section 12.10 Image Icons

44. To create an image icon for a file in `c:\book\image\icon`, use _____.

- a. `new ImageIcon("c:\book\image\icon");`
- b. `new ImageIcon('c:\book\image\icon');`
- c. `new ImageIcon("c:\\book\\image\\icon");`
- d. `new ImageIcon('c:\\book\\image\\icon');`
- e. `new ImageIcon("c:\\book\\image\\icon")` if the `c:\book` is in the class path.

Section 12.11 JButton

1. Which of the following is true?
 - a. JButton inherits from javax.swing.AbstractButton.
 - b. You can create a JButton by specifying an icon and text.
 - c. You can create a JButton by specifying an icon.
 - d. You can create a JButton by a text.
 - e. You can create a JButton using its default constructor.
2. The method _____ gets the text (or caption) of the button jbt.
 - a. jbt.text()
 - b. jbt.getText()
 - c. jbt.findText()
 - d. jbt.retrieveText().
3. The method _____ creates a ImageIcon for file c:\image\us.gif.
 - a. new ImageIcon("c:\image\us.gif");
 - b. new Icon("c:\image\us.gif");
 - c. new ImageIcon("c:\\image\\us.gif");
 - d. new Icon("c:\\image\\us.gif");
4. Which of the following are valid methods on the button jbt?
 - a. jbt.setMnemonic("A");
 - b. jbt.setMnemonic('A');
 - c. jbt.setIconTextGap(50);
 - d. jbt.setTextGap(50);
5. The method _____ specifies that the text and icon are horizontally aligned to the right in the button jbt.
 - a. jbt.setVerticalTextPosition(JButton.LEFT)
 - b. jbt.setHorizontalTextPosition(JButton.LEFT)
 - c. jbt.setHorizontalTextPosition(JButton.RIGHT)
 - d. jbt.setHorizontalAlignment(JButton.RIGHT)
 - e. jbt.setHorizontalAlignment(JButton.LEFT)
6. The method _____ specifies that the text is placed on the right of the icon in the button jbt.
 - a. jbt.setVerticalTextPosition(JButton.LEFT)
 - b. jbt.setHorizontalTextPosition(JButton.LEFT)
 - c. jbt.setHorizontalTextPosition(JButton.RIGHT)
 - d. jbt.setHorizontalAlignment(JButton.RIGHT)
 - e. jbt.setHorizontalAlignment(JButton.LEFT)
7. Analyze the following code:

```
import javax.swing.*;
import java.awt.*;

public class Test extends JFrame {
    public Test() {
        ImageIcon usIcon = new ImageIcon("image/usIcon.gif");
        JButton jbt1 = new JButton(usIcon);
        JButton jbt2 = new JButton(usIcon);

        JPanel p1 = new JPanel();
        p1.add(jbt1);

        JPanel p2 = new JPanel();
        p2.add(jbt2);

        JPanel p3 = new JPanel();
        p2.add(jbt1);
    }
}
```

```

    add(p1, BorderLayout.NORTH);
    add(p2, BorderLayout.SOUTH);
    add(p3, BorderLayout.CENTER);
}

public static void main(String[] args) {
    // Create a frame and set its properties
    JFrame frame = new Test();
    frame.setSize(200, 100);
    frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    frame.setVisible(true);
}
}

```

- a. Two buttons displayed with the same icon.
- b. Three buttons displayed with the same icon.
- c. Only jbt1 displays the icon and jbt2 does not display the icon.
- d. Two buttons displayed with different icons.

8. Analyze the following code:

```

import javax.swing.*;
import java.awt.*;

public class Test extends JFrame {
    public Test() {
        setLayout(new FlowLayout());
        add(new JButton("Java"));
        add(new JButton("Java"));
        add(new JButton("Java"));
        add(new JButton("Java"));
    }

    public static void main(String[] args) {
        // Create a frame and set its properties
        JFrame frame = new Test();
        frame.setSize(200, 100);
        frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
        frame.setVisible(true);
    }
}

```

- a. One button is displayed with the text "Java".
- b. Two buttons are displayed with the same text "Java".
- c. Three buttons are displayed with the same text "Java".
- d. Four buttons are displayed with the same text "Java".

9. Analyze the following code:

```

import javax.swing.*;
import javax.swing.border.*;
import java.awt.*;

public class Test extends JFrame {
    public Test() {
        Border border = new TitledBorder("My button");
        JButton jbt1 = new JButton("OK");
        JButton jbt2 = new JButton("Cancel");
        jbt1.setBorder(border);
        jbt2.setBorder(border);
        add(jbt1, BorderLayout.NORTH);
        add(jbt2, BorderLayout.SOUTH);
    }
}

```

```

public static void main(String[] args) {
    JFrame frame = new Test();
    frame.setSize(200, 100);
    frame.setDefaultCloseOperation(JFrame.EXIT_ON_CLOSE);
    frame.setVisible(true);
}
}

```

- a. Two buttons displayed with the same border.
- b. Two buttons displayed, but only one button has the border.
- c. The program has a compile error because you assign new TitledBorder("My button") to a variable of the Border type.
- d. The program has a runtime error because you cannot set a border on a button.

Section 12.12 JCheckBox

10. _____ checks whether the JCheckBox jchk is selected.

- a. jchk.getSelected()
- b. jchk.selected()
- c. jchk.isSelected().
- d. jchk.select()

11. Which of the following statements are true?

- a. JCheckBox inherits from javax.swing.AbstractButton.
- b. All the methods in JCheckBox are also in JButton.
- c. You can set a mnemonic for a JCheckBox.
- d. You can use an icon on JCheckBox.

Section 12.13 JRadioButton

13. Which of the following statements are true?

- a. JRadioButton inherits from javax.swing.AbstractButton.
- b. All the methods in JRadioButton are also in JButton.
- c. You can set a mnemonic for a JCheckBox.
- d. You can use an icon on JRadioButton.

14. Which of the following statements are true?

- a. You can create an instance of ButtonGroup and add radio button to the instance to group the buttons.
- b. ButtonGroup can be added to a container.
- c. To check whether a radio button jrb is selected, use jrb.isSelected().
- d. By default, when a radio button is created, the radio button is not selected.

Section 12.14 Labels

16. Which of the following statements are true?

- a. All the methods in JLabel are inherited from javax.swing.AbstractButton.
- b. You can set a mnemonic key in a JLabel.
- c. You can create a label with both text and icon.
- d. You can create a label with an icon.

17. The method _____ gets the text (or caption) of the label jlbl.

- a. jlbl.text()
- b. jlbl.getText()
- c. jlbl.findText()
- d. jlbl.retrieveText().

18. Which of the following are valid methods on the label jlbl?

- a. jlbl.setMnemonic("A");
- b. jlbl.setMnemonic('A');
- c. jlbl.setIconTextGap(50);
- d. jlbl.setTextGap(50);

19. The method _____ specifies that the text and icon are horizontally aligned to the right in the label `jlbl`.
- a. `jlbl.setVerticalTextPosition(JButton.LEFT)`
 - b. `jlbl.setHorizontalTextPosition(JButton.LEFT)`
 - c. `jlbl.setHorizontalTextPosition(JButton.RIGHT)`
 - d. `jlbl.setHorizontalAlignment(JButton.RIGHT)`
 - e. `jlbl.setHorizontalAlignment(JButton.LEFT)`
20. The method _____ specifies that the text is placed on the right of the icon in the label `jlbl`.
- a. `jlbl.setVerticalTextPosition(JButton.LEFT)`
 - b. `jlbl.setHorizontalTextPosition(JButton.LEFT)`
 - c. `jlbl.setHorizontalTextPosition(JButton.RIGHT)`
 - d. `jlbl.setHorizontalAlignment(JButton.RIGHT)`
 - e. `jlbl.setHorizontalAlignment(JButton.LEFT)`
21. The method _____ assigns the name `Result` to the `Text` of variable `jlbl`.
- a. `jlbl.setText("Result")`
 - b. `jlbl.newText("Result")`
 - c. `jlbl.text("Result")`
 - d. `jlbl.findText()`

Section 12.15 Text Fields

22. _____ can be used to enter or display a string.
- a. A label
 - b. A button
 - c. A check box
 - d. A radio button
 - e. A text field
23. Which of the following statements are true?
- a. You can specify a horizontal text alignment in a text field.
 - b. You can specify the number of columns in a text field.
 - c. You can disable editing on a text field.
 - d. You can create a text field with a specified text.
24. The method _____ gets the contents of the text field `jtf`.
- a. `jtf.getText(s)`
 - b. `jtf.getText()`
 - c. `jtf.getString()`
 - d. `jtf.findString()`